

Jiayu Liu

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EDUCATION

Purdue University

West Lafayette, IN, USA

Ph.D. in Technology

2023-2027 (Expected)

Advisor: Dr. Yi Gao

Dissertation: A Systematic Framework for Evaluating the U.S. Airports Ground Accessibility
and Temporal Reliability Using Gravity-Based Models

M.S. in Applied Statistics

2025-2026 (Expected)

M.S. in Aviation and Aerospace Management

2021-2023

Graduate Certificate in Geospatial Information Science (GIS)

2024

Beijing Jiaotong University

Beijing, China

B.E. in Traffic and Transportation (Civil Aviation Transportation)

2017-2021

CERCIFICATION & LICENSE

- ✧ Part 107 Remote Pilot Certificate
- ✧ SAS Certified Specialist: Base Programming Using SAS 9.4

PEER-REVIEWED PUBLICATIONS

- ✧ Liu, J., Kong, N. N., & Gao, Y. (2026). Assessing spatial development footprint of urban areas surrounding new airport terminal infrastructure. *Journal of Air Transport Management*.
<https://doi.org/10.1016/j.jairtraman.2025.102957>
- ✧ Gao, Y & Liu, J. (2026). Frequent Flyer Programs in Academic Research: A Literature Review of Research Progress Since 2000. *Journal of Air Transport Management*. <https://doi.org/10.1016/j.jairtraman.2025.102920>
- ✧ Liu, J., Wang, S., Dou, Z., Oluwumi, T. O., & Gao, Y. (2025). Fostering active learning through AI-integrated platforms: Student perceptions from a large enrollment professional education course. *Collegiate Aviation Review International*, 43(2), 69-99. <https://ojs.library.okstate.edu/osu/index.php/CARI/article/view/10322/9131>
- ✧ Liu, J., Wang, S., Dou, Z., & Gao, Y. (2025). Do Aging Aircraft Run Late? Evidence from the United States Domestic Flights. *Journal of the Air Transport Research Society*, 100090. <https://doi.org/10.1016/j.jatrs.2025.100090>
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2025). The fate of Akron-Canton Regional Airport following the dehubbing of Cleveland Hopkins International Airport and exodus of Southwest Airlines. *Case Studies on Transport Policy*, 101497. <https://doi.org/10.1016/j.cstp.2025.101497>
- ✧ Liu, J., Beck, C. S., & Gao, Y. (2024). Can you judge a book by its cover? An analysis of travelers' perceptions of airline logos and airlines' business models. *Journal of Air Transport Management*, 115, 102543. <https://doi.org/10.1016/j.jairtraman.2024.102543>
- ✧ Gao, Y & Liu, J. (2023). State of Airport Loyalty Programs (ACRP Synthesis 131). Washington, DC: *The National Academies Press*. <https://doi.org/10.17226/27279>

WORKING MANUSCRIPTS

- ✧ Dou, Z., Wang, S., Liu, J., & Gao, Y. Predicting Airport Passenger Arrivals from Flight Schedules: An Explainable

Machine Learning Method

- ✧ Wang, S., Liu, J., Dou, Z., & Gao, Y. Do Airlines Have Home Court Advantage? Exploring the Relationship Between Airline Dominance and Taxi-out Time

CONFERENCE ACTIVITIES

- ✧ Liu, J., & Gao, Y. (2025). Adopting AI-Integrated Online Learning Platforms in Collegiate Aviation Education [Workshop]. *The 2025 UAA Annual Collegiate Aviation Conference & Expo*. Costa Mesa, CA.
- ✧ Liu, J., Kong, N., & Gao, Y. (2024). The role of airport expansions in urban growth: The impact of new terminals on local land use [Poster & Presentation]. *Applied Geography Conference*, San Marcos, TX.
- ✧ Gao, Y. & Liu, J. (2024). Navigating the Skies of Airline Customer Loyalty: A Comprehensive Review of Airline Frequent Flyer Program Studies. *The 27th Air Transport Research Society Conference*, Lisbon, Portugal.
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2024). The Ripple Effect of Airport Dehubbing: A case study of Cleveland Hopkins International Airport and Akron Canton Airport [Presentation]. *The 27th Air Transport Research Society Conference 2024*, Lisbon, Portugal.
- ✧ Wang, S., Dou, Z., Liu, J., & Gao, Y. (2023). Spatial Determinants Behind the Seasonal Variations of Commercial Flights Across the United States [Poster & Presentation]. *The 2023 Purdue GIS Day*, West Lafayette, IN.
- ✧ Liu, J., Beck, C., & Gao, Y. (2022). Passengers' Perceptions of Airlines' Logo Design and Their Business Models. *The 25th Air Transport Research Society Conference (Hybrid)*, Antwerp, Belgium.
- ✧ Gao, Y., Liu, J., & Wang, S. (2022). Visualize Your Research Work [Workshop]. *The 2022 UAA Annual Collegiate Aviation Conference & Expo*, Colorado Springs, CO.
- ✧ Wang, S., Liu, J., Xie, H., & Gao, Y. (2022). What is Aviation Research Talking About? A Bibliographic Analysis of Three Aviation Journals during 2011-2021 [Poster & Presentation]. *Purdue University School of Aviation and Transportation Technology Poster Symposium*, West Lafayette, IN.

TEACHING EXPERIENCE

- ✧ **AT 102 - Aviation Business**, Purdue University *Fall 2024, Spring & Fall 2025*
Role: Graduate Course Instructor
 - Working as the instructor for this course, class size approximately 70-150 students vary by semester.
 - Solely responsible for designing the syllabus, developing and delivering in-person lectures, creating assessments (quizzes and exams) with clear rubrics, and managing all grading.
 - Mentoring student groups on their collaborative projects, guiding topic selection, content organization, and presentation design.
 - Integrating current industry insights and inviting guest speakers to connect foundational business concepts with practical applications in the aviation sector.
- ✧ **AT 546 - Aviation Financial Instruments and Operations**, Purdue University *Fall 2021*
Role: Graduate Teaching Assistant
 - Managed the airline simulation platform back-end (scenario setup, data resets, system checks), recorded and curated simulation data outputs (KPIs such as load factor, CASM/RASM, on-time performance) for instructor dashboards and student debriefs.
 - Supported students with technical troubleshooting, analytics tips, and interpretation of simulation results; graded assignments and short reports with rubric-based feedback.

FUNDED PROJECT

- ✧ **[ACRP] Mapping Expansion Potential: A Spatial Survey for U.S. Airports** *Principal Investigator* 2025
- ✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)
Amount: \$12,000
Period: 11/2025 - 07/2026
Tasks:
 - Define project scope, and coordinate with an industry panel (airport authorities/consultants) to gain practitioner feedback and adoption pathways.
 - Building a GIS-based suitability framework integrating land use, environmental constraints, and FAA infrastructure investment indicators.
 - Producing publishable report.

RESEARCH EXPERIENCE & PROJECT

- ✧ **The Data Mine, Purdue University** *Graduate Data Science Researcher* 2025
- ✧ **Corporate Partner:** Johnson & Johnson
Project Scope: Collaborated on Foundational Models project for early-phase clinical trials, applying Generative AI (LLMs, GANs, VAEs, Diffusion Models) and predictive modeling (XGBoost). Applied tools including Python, R, and SQL in large-scale biomedical data analysis.
Tasks:
 - Building data pipelines in Python for large, de-identified biomedical datasets; performing feature selection, cross-validation, and survival analysis.
 - Delivering reproducible notebooks, documentation, and technical reports; collaborating with a cross-functional team on review cycles and presentation of findings.
- ✧ **[ACRP Synthesis 131] State of Airport Loyalty Programs** *Graduate Research Assistant* 2022
- ✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)
Project Scope: Investigating customer loyalty programs in different domains, especially major airports in the US. Providing a comprehensive overview of the current status of airport loyalty programs.
Tasks:
 - Conducting systematic literature review and industry scan on airport loyalty practices; compiling comparative tables of program structures and incentives.
 - Assisting with survey design and outreach to airports; cleaning and summarizing responses; drafting synthesis sections on trends, gaps, and implementation challenges.
 - Supporting figure/table production and citation management; coordinating edits with PI/co-authors to align with TRB/NAS editorial standards.
- ✧ **[ACRP 01-41] Airport Blockchain Implementation Guidebook** *Graduate Research Assistant* 2021
- ✧ **Sponsor:** Airport Cooperative Research Program (ACRP), Transportation Research Board (TRB), National Academies of Sciences, Engineering, and Medicine (NASEM)
Project Scope: Collecting and summarizing blockchain and associated technologies implementation in the aviation industry to provide business solutions and operational efficiencies for airport decision-makers and stakeholders.
Tasks:
 - Conducting systematic online scan of blockchain use cases in aviation contexts, including identity and credentialing, asset tracking, maintenance records, supply chain, and payments.
 - Produced synthesized summaries, decision checklists, and implementation roadmaps.

ACADEMIA SERVICE & ACTIVITIES

- ✧ **Reviewer**

- Journal of Air Transport Management (JATM)
- Collegiate Aviation Review-International (CAR-i)
- Reliability Engineering & System Safety

✧ **Member**

- American Association of Airport Executives (AAAE)
- American Institute of Aeronautics and Astronautics (AIAA)
- University Aviation Association (UAA)
- Air Transport Research Society (ATRS)

INTERNSHIP

- | | | |
|---|--|-------------|
| ✧ China Mining Logistics Holdings Co., Ltd | <i>International Logistics Analyst</i> | <i>2020</i> |
| – Assisted with freight forwarding, customs declaration and inspection, filling out, photocopying and binding various documents | | |
| ✧ Citic-Prudential Fund Management Co., Ltd. | <i>Financial Data Analyst</i> | <i>2019</i> |
| – Organizing the records of the brokerage roadshow/fund research | | |
| – Collecting and analyzing information on the macroeconomy, capital market, and public offering fund | | |
| – Cleaning up and processing data | | |

SKILLS

- ✧ Statistical Programming/Markup Languages: R, Python, SAS, SPSS, LATEX
- ✧ Languages: English, Mandarin
- ✧ Applications: Microsoft Office, Tableau, ArcGIS Pro, AutoCAD, VOSviewer, Adobe Creative Suite, Premiere Pro
- ✧ Music: Violin Performing and Teaching, Piano, Orchestra Performing

HONORS & AWARDS

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| ✧ Graduate Ross Fellowship, Purdue University | <i>2023-2027</i> |
| ✧ Industry Advisory Board Research Poster Symposium | <i>2022</i> |
| ✧ First Prize for 16th National Competition of Transport Science and Technology for University Student | <i>2021</i> |
| ✧ Purdue Bands & Orchestras Scholarship | <i>2021</i> |
| ✧ China Undergraduate Mathematical Contest in Modeling | <i>2020</i> |

COMMUNITY SERVICES & ACTIVITIES

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|--|---|--------------------|
| ✧ Aviation Graduate Council (AGC), Purdue University | <i>Member</i> | <i>Fall 2022</i> |
| – Present for research roundtable | | |
| ✧ Concert Orchestra, Purdue University | <i>Concertmaster</i> | <i>Fall 2022</i> |
| – Organize orchestra rehearsals and concerts | | |
| – First violin sectional rehearsal leader | | |
| – Perform violin solo part | | |
| ✧ Festival Chorus and Orchestra, Purdue & Ball State University | <i>Concertmaster</i> | <i>Summer 2022</i> |
| – Organize rehearsals during summer tour to Europe | | |
| – Perform violin solo part | | |
| ✧ Philharmonic Orchestra, Purdue University | <i>Associate Concertmaster</i> | <i>2021</i> |
| – Perform part of the violin solo works | | |
| ✧ Suzuki Association of the Americas | <i>Registered Suzuki Violin Teacher</i> | <i>2022</i> |
| – Core unites training | | |